

Talha Jobayer Zihan

jobayertalha2020@gmail.com • +8801721577792 • [linkedin.com/in/talha-jobayer](https://www.linkedin.com/in/talha-jobayer) • github.com/jobayertalha • [Portfolio](#)

SUMMARY

AI/ML engineer, CSE graduate (RUET) with two IEEE-published NLP papers and hands-on experience shipping ML systems end to end: RAG pipelines, transformer fine-tuning, and PyTorch models served through FastAPI and React apps.

EXPERIENCE

Research Intern, ELITE Research Lab Jun 2026 – Present

- R&D in **AI and Machine Learning**: literature review, model development, and experimentation.

Industrial Training, Netro Systems Rajshahi | 2 Weeks

- Used industry-standard development tools and collaborated with developers on real projects.

Industrial Attachment, Frontier Semiconductor BD Ltd Dhaka | 2 Weeks

- Observed semiconductor fabrication processes and quality control.

PROJECTS

CareerVector – RAG-Based CV Analyzer [GitHub](#)

- Matches CVs to job descriptions with sub-second retrieval; cut query latency by **80%** and lifted top-1 precision by **35%** over a TF-IDF baseline.
- **Stack**: LangChain, FAISS (IVF), OpenAI Embeddings, Python.

Cross-Domain NIDS – Network Intrusion Detection Platform [GitHub](#)

- Detects attacks in PCAP/PCAPNG traffic across different network domains using transfer learning (**Lopez17CNN**, **BiC**, **TDNR**), with an analyst dashboard for uploads, threats, and alerts.
- **Stack**: PyTorch, FastAPI, React, TypeScript, Vite; deployed on Vercel and Render.

Bangla Sarcasm Detection using Transformers [Kaggle](#)

- Classifies sarcasm in Bangla text at **88% F1** (10k balanced samples); regularization cut overfitting by **15%** and converged in 3 fewer epochs.
- **Stack**: BanglaBERT, Mixup, FGM adversarial training, R-Drop, OneCycleLR.

ResearchFlow – Research Task Management Platform [GitHub](#)

- Tracks multiple papers, tasks, conferences, and deadlines in separate workflows, with AI-generated progress insights and next-task suggestions.
- **Stack**: React, Vite, FastAPI, PostgreSQL, Supabase, Groq AI; Vercel and Cloudflare Workers (Hyperdrive).

Image Classifier using CNN TensorFlow / Keras

- Classifies handwritten digits at **95% test accuracy** on 60k MNIST samples.
- **Stack**: TensorFlow, Keras, 3-layer CNN with batch normalization.

Car Rental System Web App [GitHub](#)

- Manages car bookings with real-time date validation, booking history, secure login, and an admin panel.
- **Stack**: PHP, MySQL, HTML/CSS, JavaScript.

RESEARCH WORK

Fast Gradient Method (FGM) Fine-Tuning for Bengali Sarcasm Detection [IEEE Xplore](#)

- Proposed FGM adversarial fine-tuning on BanglaBERT; presented at ICECTE 2026 (Certificate of Contribution).

Mental Health Text Classification via Data Augmentation and Transformer Ensemble [IEEE Xplore](#)

- Improved classification accuracy using data augmentation and transformer model ensembling.

TECHNICAL SKILLS

AI / ML: PyTorch, TensorFlow, Transformers, BanglaBERT, CNNs, RAG, LangChain, FAISS, NumPy, Pandas

Programming: Python, C++, C, Java, SQL, PHP

Web / Backend: FastAPI, React, Next.js, TypeScript, Vite, Tailwind CSS, HTML/CSS, JavaScript

Tools: Git, GitHub, Docker, Linux, Kaggle, MySQL, Arduino, Vercel, Render

EDUCATION

Rajshahi University of Engineering and Technology (RUET) 2022 – 2026

B.Sc. in Computer Science and Engineering CGPA: 3.64 / 4.00

New Govt. Degree College, Rajshahi 2018 – 2020

Higher Secondary Certificate (HSC) GPA: 5.00 / 5.00

CERTIFICATIONS

[Machine Learning in Production](#) (DeepLearning.AI) • [NLP in TensorFlow](#) (DeepLearning.AI) • [Deep Learning & Reinforcement Learning](#) (IBM) • [Machine Learning with Python](#) (IBM) • [Intro to Neural Networks and PyTorch](#) (IBM) • [Agents Course](#) (Hugging Face) • [PHP \(Laravel\) Training](#) – RUET CSE, EDGE (80 hrs)

LEADERSHIP & VOLUNTEERING

Officer – Marketing, TEDxRUET 2024 – 2025

Co-organized a TEDx event; managed marketing outreach and audience engagement.